

RQ Answers

omitted

2024-04-05

Data overview

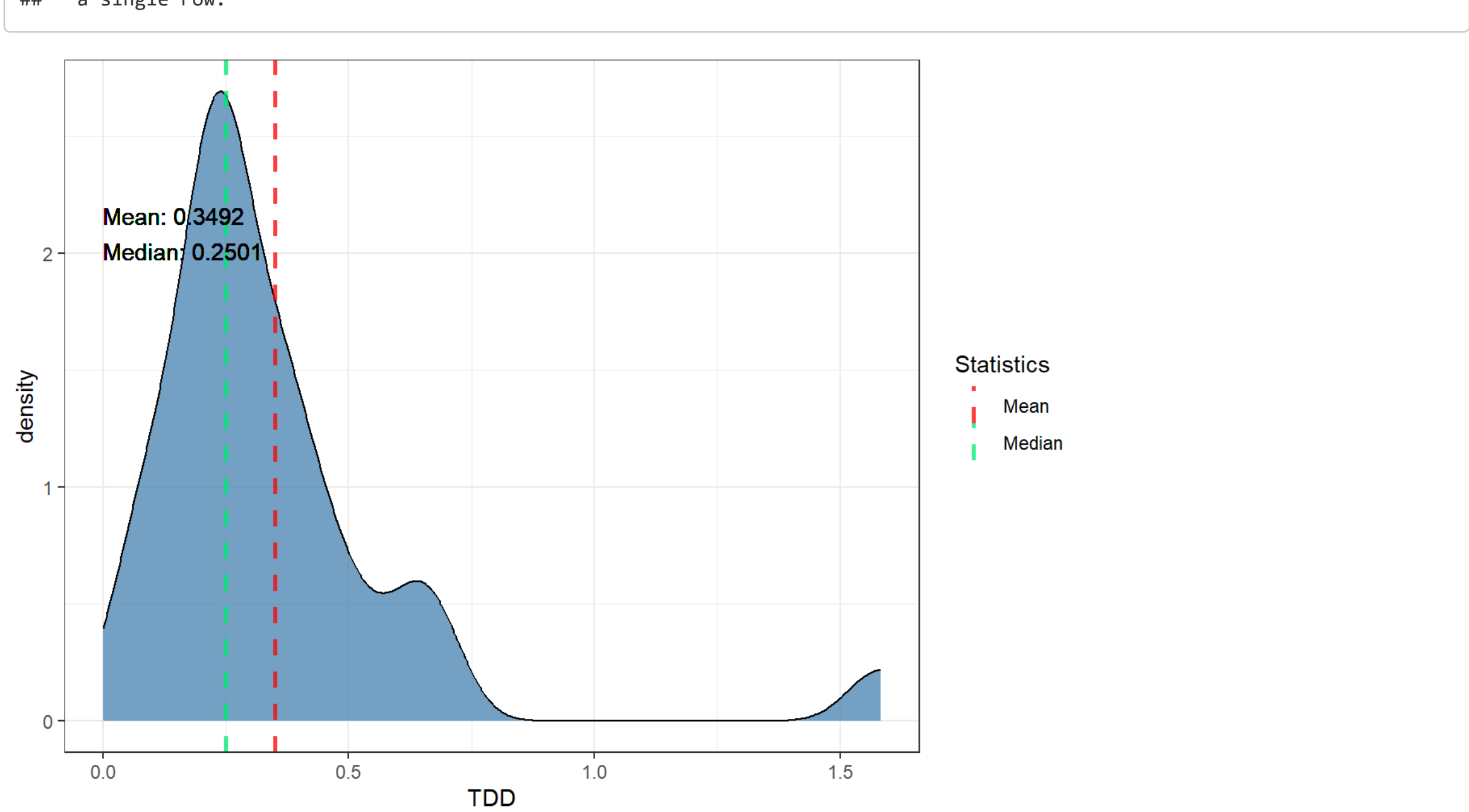
glimpse(all_data)				
## Rows: 1,140				
## Columns: 11				
## \$ repo	<chr>	"zookeeper", "zookeeper", "zookeeper", "zookeeper"...		
## \$ pr_number	<int>	1690, 1690, 1690, 1690, 1690, 1690, 1690, 16...		
## \$ rule	<chr>	"java:S1192", "java:S115", "java:S2293", "java:S11...		
## \$ severity	<chr>	"CRITICAL", "CRITICAL", "MINOR", "MINOR", "MINOR",...		
## \$ file	<chr>	"zookeeper-server/src/main/java/org/apache/zookeep...		
## \$ type	<chr>	"CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...		
## \$ status	<chr>	"OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...		
## \$ debt	<int>	8, 2, 1, 2, 5, 8, 40, 18, 10, 44, 15, 5, 15, 11, 1...		
## \$ ncloc_affected_file	<int>	821, 821, 821, 821, 821, 821, 821, 821, 821, 821, ...		
## \$ complexity	<int>	146, 146, 146, 146, 146, 146, 146, 146, 146, 146, ...		
## \$ origin	<chr>	"NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTI...		

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd)), : All aesthetics have length 1, but the data has 28 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd)), : All aesthetics have length 1, but the data has 28 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

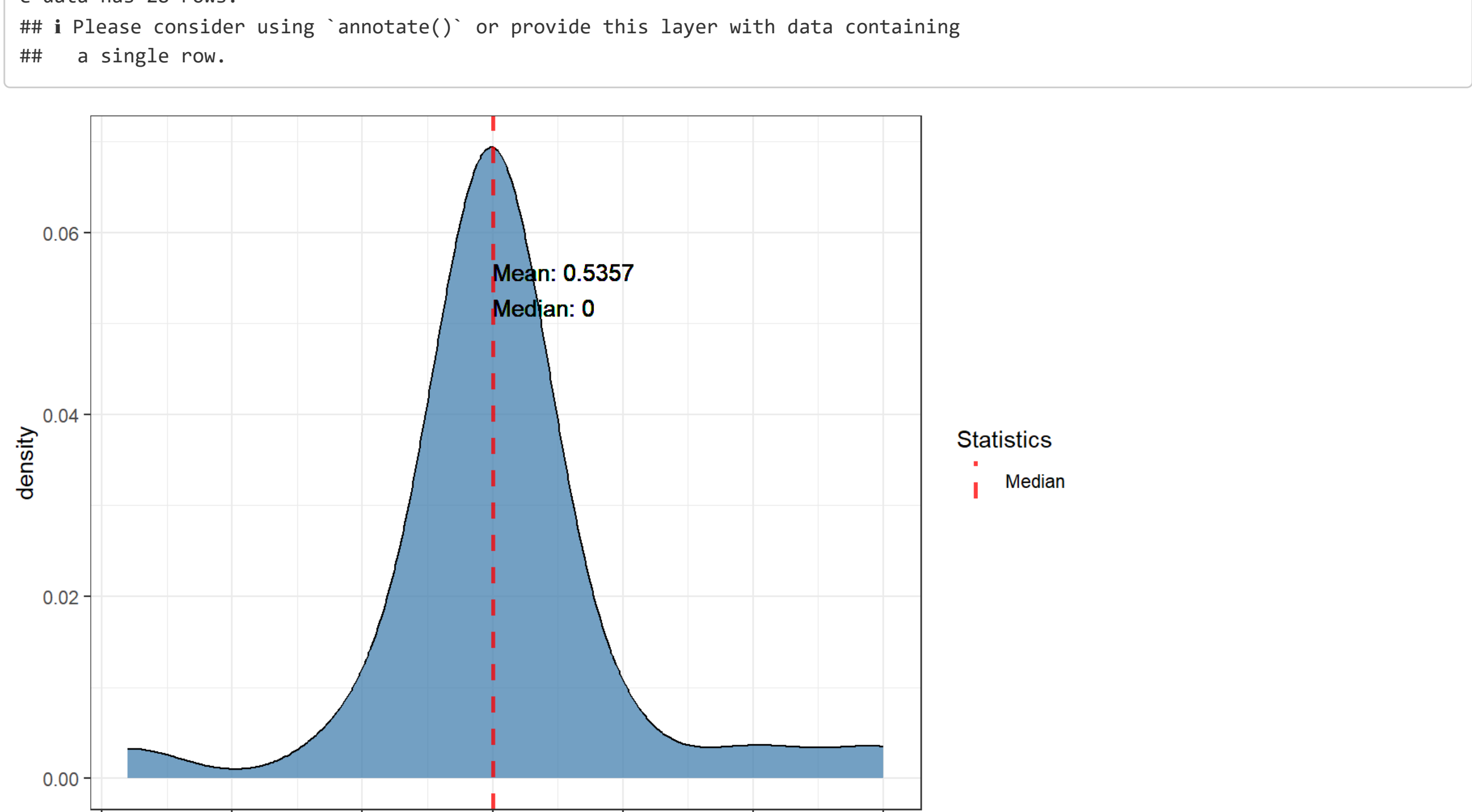


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv)), : All aesthetics have length 1, but the data has 28 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv)), : All aesthetics have length 1, but the data has 28 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 x 4
##   repo      tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 zookeeper              14.3              71.4              14.3
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 x 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##   <dbl>                <dbl>          <dbl>          <dbl>
## 1              14.3              71.4              14.3
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 x 3
##   repo      preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 zookeeper              78.6              21.4
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##   <dbl>                <dbl>          <dbl>          <dbl>
## 1              78.6              21.4
```

Percentages for new TDV by repo

```
## # A tibble: 1 x 3
##   # Groups:   repo [1]
##   repo      new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 zookeeper              85.7              14.3
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##   <dbl>                <dbl>          <dbl>          <dbl>
## 1              85.7              14.3
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 x 2				
## # Groups: rank [10]				
## rank zookeeper				
## <int> <chr>				
## 1	1	java:S2293	-	12
## 2	2	java:S3776	-	3
## 3	3	java:S5993	-	2
## 4	4	java:S1104	-	2
## 5	5	java:S3824	-	2
## 6	6	java:S2095	-	1
## 7	7	java:S1130	-	1
## 8	8	java:S2184	-	1
## 9	9	java:S1117	-	1
## 10	10	java:S1161	-	1

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 10 x 2		
## rule position_score		
## <chr> <dbl>		
## 1	java:S2293	122
## 2	java:S3776	123
## 3	java:S1104	124
## 4	java:S3824	125
## 5	java:S5993	126
## 6	java:S1117	127
## 7	java:S1130	128
## 8	java:S1135	129
## 9	java:S1161	130
## 10	java:S2065	131

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 x 2				
## # Groups: rank [10]				
## rank zookeeper				
## <int> <chr>				
## 1	1	java:S106	-	98
## 2	2	java:S3776	-	78
## 3	3	java:S115	-	78
## 4	4	java:S2142	-	69
## 5	5	java:S1104	-	62
## 6	6	java:S1117	-	49
## 7	7	java:S1192	-	46
## 8	8	java:S1133	-	39
## 9	9	java:S2293	-	39
## 10	10	java:S1141	-	33

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 x 2		
## rule position_score		
## <chr> <dbl>		
## 1	java:S106	122
## 2	java:S115	123
## 3	java:S3776	124
## 4	java:S2142	125
## 5	java:S1104	126
## 6	java:S1117	127
## 7	java:S1192	128
## 8	java:S1133	129
## 9	java:S2293	130
## 10	java:S1141	131

Outliers

Higher TDD

Outliers

Higher TDD

## # A tibble: 28 x 5					
##	repo	pr_number	total_debt	total_ncloc	tdd
##	<chr>	<int>	<int>	<int>	<dbl>
## 1	zookeeper	1994	821	519	1.58
## 2	zookeeper	1992	815	1217	0.670
## 3	zookeeper	2099	177	266	0.665
## 4	zookeeper	1915	223	374	0.596
## 5	zookeeper	2081	90	184	0.489

Higher TDV

```
## # A tibble: 28 x 3
```

## pr_number repo tdv			
## <int> <chr> <int>			
## 1	2001	zookeeper	-
## 2	2099	zookeeper	-
## 3	1690	zookeeper	-
## 4	2114	zookeeper	-
## 5	1697	zookeeper	-
## 6	1799	zookeeper	-
## 7	1954	zookeeper	-
## 8	1985	zookeeper	-
## 9	1992	zookeeper	-
## 10	1994	zookeeper	-
## # i 18 more rows			

Lower TDV

##	pr_number	repo	tdv
##	<int>	<chr>	<int>
## 1	2001	zookeeper	30
## 2	2099	zookeeper	20

Extra

Number of PRs with pre-existing TD

## n	
## 1	28

Number of java:S1192 issues that affect test code

## n	
## 1	0

Number of java:S1192 issues that affect not test code

## n	
## 1	46

Summary NCLOC

## ncloc	
## Min.	: 64.0
## 1st Qu.:	385.2
## Median	:1169.5
## Mean	:1332.0
## 3rd Qu.:	1807.8
## Max.	:6480.0